Properties of real numbers

1

a True **b** True **c** True **d** True

e False **f** True **g** True **h** True

i True **j** True **k** False

2 False

3 False

4

a True **b** True **c** True **d** True

5

a $0 + 9m = 9m$ **b** $1 \times 4b = 4b$

c $3d - 0 = 3d$ **d** $a \times 19 = 19 \times a$

e $d + \frac{1}{7} = \frac{1}{7} + d$

6

a $5\,(s + 6) = 5\,(6 + s)$ **b** $8\,(4 + m) = 8(m + 4)$

7

a No **b** No **c** Yes **d** No

8 $\frac{1}{2x}$

9

a Commutative property **b** Commutative property

c Commutative property **d** Associative property

e Associative property **f** Associative property

g Identity property **h** Inverse property

i Inverse property **j** Inverse property

k Identity property **l** Identity property

m Commutative property **n** Commutative property

o Commutative property **p** Associative property

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a Commutative property

b Associative property

c Inverse property

d Associative property

e Associative property

f Associative property

g Commutative property

h Associative property

i Identity property

11 Multiplication property of equality

12

a Yes **b** Yes **c** Yes

13

a

i Commutative property of addition **ii** $-7 + 8 = 8 + (-7)$

b

i Commutative property of multiplication **ii** $-6 \times 5 = 5 \times (-6)$

c

i Associative property of addition **ii** $(9 + 7) + 6 = 9 + (7 + 6)$

d

i Associative property of multiplication **ii** $6 \times (9 \times 7) = (6 \times 9) \times 7$

14

a $3 + x$ **b** $q \times 4$ **c** $d \times g$

15

a $2 \times (a \times b) = (2 \times a) \times b$

b $(ab)\,c = a\,(bc)$

c $9 + (a + b) = (9 + a) + b$

16 $8\,(5h) = (8 \times 5)\,h$

$= 40h$

17 $3 \times 6 + 3 \times 5$

18 The sum of a real number and its additive inverse is equal to 0, the additive identity.

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19 False, because the commutative property allows us to swap added terms but not the coefficients of those terms.

20

a Distributive property

b Commutative property of addition

c Associative property of addition

d Commutative property of addition

Additional questions

21

a Commutative property, reordering

b Commutative property, reordering

c Reordering

d Associative property, commutative property, reordering

e Distributive property

f Associative property, commutative property, reordering

22

a True **b** False **c** False **d** True

23

a 75

b 58

24

a $27 + 13$

b 88

25 260

26 79

27

a Yes

b Reordering

28 2

29

a 26

b 12

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30 11

31

a Yes

b Reordering

32

a 4×5

b $4 \times 13 \times 5 = 4 \times 5 \times 13$ or $4 \times 13 \times 5 = 5 \times 4 \times 13$

33 340

34

a 3

b 20

c Distributive property

d 805

35

a 30

b 1

c Distributive property

d 754

36

a 260

b 26

c 286

37

a 320

b 32

c 288

38

a 8982

b 7014

c 1364

d 2472

39 4

40

a 28

b 28

c Distributive property

d 24

41

a 21

b 7

c Distributive property

d 17

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a 202 **b** 99 **c** 87 **d** 45

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